

2. Implement of error correction code (like CRC).

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void xorOperation(char *dividend, char *divisor, int len){
    for (int i = 1; i < len; i++){
        if(dividend[i] == divisor[i])
            dividend[i] = '0';
        else
            dividend[i] = '1';
    }
}

void crcDivision(char* data, char* generator, char* remainder){
    int dataLen = strlen(data);
    int genLen = strlen(generator);
    char temp[100];
    strncpy(temp, data, genLen);
    temp[genLen] = '\0';

    int cur = genLen;

    while(cur < dataLen){
        if(temp[0] == '1')
            xorOperation(temp, generator, genLen);

        memmove(temp, temp + 1, genLen - 1);
        temp[genLen - 1] = data[cur];
        cur++;
    }
    if(temp[0] == '1')
```

```

        xorOperation(temp, generator, genLen);
strcpy(remainder, temp + 1);
}

int main(){
    char data[100], generator[100];
    char appendedData[100], remainder[100];

    printf("Enter data bits: ");
    scanf("%s", data);

    printf("Enter generator bits: ");
    scanf("%s", generator);

    int dataLen = strlen(data);
    int genLen = strlen(generator);

    strcpy(appendedData, data);

    // Append zeros
    for(int i = 0; i < genLen - 1; i++){
        appendedData[dataLen + i] = '0';
    }
    appendedData[dataLen + genLen - 1] = '\0';

    crcDivision(appendedData, generator, remainder);
    printf("CRC Remainder: %s\n", remainder);

    char finalData[100];
    strcpy(finalData, data);
    strcat(finalData, remainder);

    printf("Transmitted codeword: %s\n", finalData);

```

```

char received[100], checkRemainder[100];
printf("Enter received codeword: ");
scanf("%s", received);

crcDivision(received, generator, checkRemainder);

if(strspn(checkRemainder, "0") == strlen(checkRemainder))
    printf("No error detected\n");
else
    printf("Error detected\n");

return 0;
}

```

3. Simulate the transmission of ping messages over a network topology consisting of 6 nodes.

```

# Create simulator
set ns [new Simulator]

# NAM and Trace files
set nf [open p3.nam w]
$ns namtrace-all $nf
set nd [open p3.tr w]
$ns trace-all $nd

# Finish procedure
proc finish {} {
    global ns nf nd
    $ns flush-trace
    close $nf
}

```

```

        close $nd
        exec nam p3.nam &
        exit 0
    }

# Create nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]

# Create links
$ns duplex-link $n1 $n0 1Mb 12ms DropTail
$ns duplex-link $n2 $n0 1Mb 10ms DropTail
$ns duplex-link $n3 $n0 1Mb 10ms DropTail
$ns duplex-link $n4 $n0 1Mb 10ms DropTail
$ns duplex-link $n5 $n0 1Mb 10ms DropTail
$ns duplex-link $n6 $n0 1Mb 11ms DropTail

# Orientation for NAM visualization
$ns duplex-link-op $n1 $n0 orient right-down
$ns duplex-link-op $n2 $n0 orient left-up
$ns duplex-link-op $n3 $n0 orient right
$ns duplex-link-op $n4 $n0 orient right-up
$ns duplex-link-op $n5 $n0 orient left-down
$ns duplex-link-op $n6 $n0 orient left

# Ping receive procedure
Agent/Ping instproc recv {from rtt} {
    $self instvar node_

```

```
    puts "node[$node_id] receive ping answer from $from with round-trip-time  
$rtt ms."  
}
```

```
# Create ping agents
```

```
set p1 [new Agent/Ping]  
set p2 [new Agent/Ping]  
set p3 [new Agent/Ping]  
set p4 [new Agent/Ping]  
set p5 [new Agent/Ping]  
set p6 [new Agent/Ping]
```

```
# Labels for nodes
```

```
$n0 label "center"  
$n1 label "ping1"  
$n2 label "ping2"  
$n3 label "ping3"  
$n4 label "ping4"  
$n5 label "ping5"  
$n6 label "ping6"
```

```
# Colors
```

```
$ns color 1 red  
$ns color 2 blue  
$ns color 3 green  
$ns color 4 orange  
$ns color 5 purple
```

```
# Attach agents to nodes
```

```
$ns attach-agent $n1 $p1  
$ns attach-agent $n2 $p2  
$ns attach-agent $n3 $p3  
$ns attach-agent $n4 $p4
```

```
$ns attach-agent $n5 $p5
```

```
$ns attach-agent $n6 $p6
```

```
# Queue limits on center node
```

```
$ns queue-limit $n0 $n4 3
```

```
$ns queue-limit $n0 $n5 2
```

```
$ns queue-limit $n0 $n6 2
```

```
# Set class for agents
```

```
$p1 set class_ 1
```

```
$p2 set class_ 2
```

```
$p3 set class_ 3
```

```
$p4 set class_ 4
```

```
$p5 set class_ 5
```

```
$p6 set class_ 1
```

```
# Connect ping agents to the center node (so they can send/receive)
```

```
$ns connect $p1 $p2
```

```
$ns connect $p2 $p3
```

```
$ns connect $p3 $p4
```

```
$ns connect $p4 $p5
```

```
$ns connect $p5 $p6
```

```
$ns connect $p6 $p1
```

```
# Schedule ping sends
```

```
$ns at 0.1 "$p1 send"
```

```
$ns at 0.3 "$p2 send"
```

```
$ns at 0.5 "$p3 send"
```

```
$ns at 1.0 "$p4 send"
```

```
$ns at 1.2 "$p5 send"
```

```
$ns at 1.4 "$p6 send"
```

```
# Finish simulation
```

```
$ns at 2.0 "finish"
```

```
# Run the simulation
```

```
$ns run
```

4. Simulate three nodes point- to - point network with duplex links between them. Set the queue size and vary the bandwidth and find the number of packets dropped.

```
# Create Simulator
```

```
set ns [new Simulator]
```

```
$ns color 1 Red
```

```
# Trace Files
```

```
set nt [open p4.tr w]
```

```
$ns trace-all $nt
```

```
set na [open p4.nam w]
```

```
$ns namtrace-all $na
```

```
# Create Nodes
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
# Node Colors
```

```
$n0 color Blue
```

```
$n1 color Pink
```

```
$n2 color Purple
```

```
# Node Labels

$ns at 0.0 "$n0 label TCP-Source"
$ns at 0.0 "$n1 label Center-node"
$ns at 0.0 "$n2 label TCP-Sink"


# Links

$ns duplex-link $n0 $n1 10Mb 1ms DropTail
$ns queue-limit $n0 $n1 10

$ns duplex-link $n1 $n2 80Mb 1ms DropTail
$ns queue-limit $n1 $n2 10


# TCP Agent

set tcp0 [new Agent/TCP]
$tcp0 set class_ 1
$ns attach-agent $n0 $tcp0


# Sink

set sink [new Agent/TCPSink]
$ns attach-agent $n2 $sink


# Connect Agents

$ns connect $tcp0 $sink


# FTP Application

set ftp0 [new Application/FTP]
$ftp0 set packetSize_ 500
$ftp0 set interval_ 0.005
$ftp0 attach-agent $tcp0


# Finish Procedure

proc End {} {
    global ns na nt
```



```
$ns flush-trace
close $na
close $nt
exec nam p4.nam &
exit 0
}
```

```
# Start and Stop
$ns at 0.0 "$ftp0 start"
$ns at 20.0 "End"
```

```
# Run Simulation
$ns run
```

```
----- p4.awk-----
BEGIN {
    Count = 0
}
{
    if ($1 == "d") {
        Count++
    }
}
END {
    printf("\nNumber of packets dropped is: %d\n", Count)
}
```

5. Simulate five nodes point- to - point network with duplex links between them. Set the queue size and vary the bandwidth and find the number of packets sent with different types of traffic.

```
# Create a Simulator object
```

```
set ns [new Simulator]
```

```
# Define colors for NAM
```

```
$ns color 1 Blue
```

```
$ns color 2 Red
```

```
# Open the Trace file and NAM trace file
```

```
set tracefile [open p5.tr w]
```

```
$ns trace-all $tracefile
```

```
set namfile [open p5.nam w]
```

```
$ns namtrace-all $namfile
```

```
# Create five nodes
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
# Create Duplex Links
```

```
$ns duplex-link $n0 $n1 2Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n2 2Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n3 2Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n4 2Mb 10ms DropTail
```

```
# Set Queue Limit (Size) to 20
```

```
$ns queue-limit $n0 $n1 20
```

```
$ns queue-limit $n0 $n2 20
```

```
$ns queue-limit $n0 $n3 20
```

```
$ns queue-limit $n0 $n4 20

# --- Setup UDP Traffic (n1 to n3) ---
set udp [new Agent/UDP]
$ns attach-agent $n1 $udp
set null [new Agent/Null]
$ns attach-agent $n3 $null
$ns connect $udp $null
$udp set fid_ 1

set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set packetSize_ 1000
$cbr set rate_ 1Mb

# --- Setup TCP Traffic (n2 to n4) ---
set tcp [new Agent/TCP]
$ns attach-agent $n2 $tcp
set sink [new Agent/TCPSink]
$ns attach-agent $n4 $sink
$ns connect $tcp $sink
$tcp set fid_ 2

set ftp [new Application/FTP]
$ftp attach-agent $tcp

# Define a 'finish' procedure
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam out.nam &
```

```
        exit 0
    }

# Schedule events
$ns at 0.5 "$cbr start"
$ns at 1.0 "$ftp start"
$ns at 4.0 "$ftp stop"
$ns at 4.5 "$cbr stop"
$ns at 5.0 "finish"

# Run the simulation
$ns run
```

P5.awk

```
BEGIN { cbr=0; tcp=0; }
{
    if ($1 == "+")
    {
        if($5 == "cbr") cbr++;
        if($5 == "tcp") tcp++;
    }
}
END {
    print "UDP Packets Sent = ", cbr;
    print "TCP Packets Sent = ", tcp;
    print "Total Packets Sent = ", cbr + tcp;
}
```

6. Implement distance vector algorithm to find the suitable path for transmission between sender and receiver.

```
set ns [new Simulator]
```

```
$ns rtproto DV
```

```
$ns color 1 Blue
```

```
$ns color 2 Red
```

```
set nt [open 6.tr w]
```

```
$ns trace-all $nt
```

```
set na [open 6.nam w]
```

```
$ns namtrace-all $na
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
$n0 color Blue
```

```
$n5 color Red
```

```
$n0 label "Source"
```

```
$n5 label "Receiver"
```

```
$n1 label "R1"
```

```
$n2 label "R2"
```

```
$n3 label "R3"
```

```
$n4 label "R4"
```

```
$ns duplex-link $n0 $n1 10Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n2 10Mb 10ms DropTail
```

```
$ns duplex-link $n1 $n3 10Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n4 10Mb 10ms DropTail
$ns duplex-link $n2 $n4 10Mb 10ms DropTail
$ns duplex-link $n4 $n5 10Mb 10ms DropTail
```

```
set tcp [new Agent/TCP]
$tcp set class_ 1
$ns attach-agent $n0 $tcp
set sink [new Agent/TCPSink]
$ns attach-agent $n5 $sink
$ns connect $tcp $sink
```

```
set ftp [new Application/FTP]
$ftp attach-agent $tcp
```

```
set udp [new Agent/UDP]
$udp set class_ 2
$ns attach-agent $n0 $udp
set null [new Agent/Null]
$ns attach-agent $n5 $null
$ns connect $udp $null
```

```
set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
```

```
$ns rtmodel-at 1.5 down $n0 $n2
$ns rtmodel-at 3.0 up $n0 $n2
```

```
proc finish {} {
    global ns nt na
    $ns flush-trace
    close $nt
    close $na
    exec nam 6.nam &
```

```
    exit 0
}
$ns at 0.1 "$ftp start"
$ns at 0.1 "$cbr start"
$ns at 4.5 "$ftp stop"
$ns at 4.5 "$cbr stop"
$ns at 5.0 "finish"

$ns run
```

7. Simulation of Link State Routing algorithm.

```
set ns [new Simulator]

$ns rtproto LS

$ns color 1 Blue
$ns color 2 Red

set nt [open 7.tr w]
$ns trace-all $nt
set na [open 7.nam w]
$ns namtrace-all $na

set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]

$n0 color Blue
```

```
$n5 color Red
```

```
$n0 label "N0(Source)"
```

```
$n5 label "N5(Dest)"
```

```
$n1 label "N1"
```

```
$n2 label "N2"
```

```
$n3 label "N3"
```

```
$n4 label "N4"
```

```
$ns duplex-link $n0 $n1 10Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n2 10Mb 10ms DropTail
```

```
$ns duplex-link $n1 $n3 10Mb 10ms DropTail
```

```
$ns duplex-link $n2 $n4 10Mb 10ms DropTail
```

```
$ns duplex-link $n4 $n5 10Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n5 10Mb 10ms DropTail
```

```
$ns duplex-link $n1 $n4 10Mb 10ms DropTail
```

```
$ns duplex-link $n2 $n3 10Mb 10ms DropTail
```

```
set tcp [new Agent/TCP]
```

```
$tcp set class_ 1
```

```
$ns attach-agent $n0 $tcp
```

```
set sink [new Agent/TCPSink]
```

```
$ns attach-agent $n5 $sink
```

```
$ns connect $tcp $sink
```

```
set ftp [new Application/FTP]
```

```
$ftp attach-agent $tcp
```

```
set udp [new Agent/UDP]
```

```
$udp set class_ 2
```

```
$ns attach-agent $n0 $udp
```

```
set null [new Agent/Null]
```

```
$ns attach-agent $n5 $null
```



```
$ns connect $udp $null
```

```
set cbr [new Application/Traffic/CBR]
```

```
$cbr attach-agent $udp
```

```
$ns rtmodel-at 1.5 down $n1 $n3
```

```
$ns rtmodel-at 2.5 up $n1 $n3
```

```
proc finish {} {
```

```
    global ns nt na
```

```
    $ns flush-trace
```

```
    close $nt
```

```
    close $na
```

```
    exec nam 7.nam &
```

```
    exit 0
```

```
}
```

```
$ns at 0.1 "$ftp start"
```

```
$ns at 0.1 "$cbr start"
```

```
$ns at 4.5 "$ftp stop"
```

```
$ns at 4.5 "$cbr stop"
```

```
$ns at 5.0 "finish"
```

```
$ns run
```

8. Simulation of Routing Information Protocol.

```
set ns [new Simulator]
```

```
$ns rtproto DV
```

```
$ns color 1 Black
```

```
set nt [open 8.tr w]
```

```
$ns trace-all $nt
```

```
set na [open 8.nam w]
```

```
$ns namtrace-all $na
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
set n6 [$ns node]
```

```
$n0 label "Source"
```

```
$n3 label "Reciever"
```

```
$n1 label "R1"
```

```
$n2 label "R2"
```

```
$n4 label "R4"
```

```
$n5 label "R5"
```

```
$n6 label "R6"
```

```
$ns duplex-link $n0 $n1 10Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n6 10Mb 10ms DropTail
```

```
$ns duplex-link $n1 $n2 10Mb 10ms DropTail
```

```
$ns duplex-link $n2 $n3 10Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n4 10Mb 10ms DropTail
$ns duplex-link $n4 $n5 10Mb 10ms DropTail
$ns duplex-link $n6 $n5 10Mb 10ms DropTail
```

```
set tcp [new Agent/TCP]
$tcp set class_ 1
$ns attach-agent $n0 $tcp
set sink [new Agent/TCPSink]
$ns attach-agent $n3 $sink
$ns connect $tcp $sink
```

```
set ftp [new Application/FTP]
$ftp attach-agent $tcp
```

```
set udp [new Agent/UDP]
$udp set class_ 2
$ns attach-agent $n0 $udp
set null [new Agent/Null]
$ns attach-agent $n3 $null
$ns connect $udp $null
```

```
set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
```

```
$ns rtmodel-at 1.5 down $n0 $n1
$ns rtmodel-at 3.0 up $n0 $n1
```

```
proc finish {} {
    global ns nt na
    $ns flush-trace
    close $nt
    close $na
    exec nam 8.nam &
```

```
        exit 0
    }
    $ns at 0.1 "$ftp start"
    $ns at 0.1 "$cbr start"
    $ns at 4.5 "$ftp stop"
    $ns at 4.5 "$cbr stop"
    $ns at 5.0 "finish"

    $ns run
```

9. Simulate an Ethernet LAN using n nodes, change error rate and data rate and compare throughput.

```
set ns [new Simulator]

$ns color 1 Blue

set nt [open 9.tr w]
$ns trace-all $nt
set na [open 9.nam w]
$ns namtrace-all $na

set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
set n7 [$ns node]
```

\$n0 color Red

\$n1 color Red

\$n2 color Red

\$n3 color Red

\$n4 color Blue

\$n5 color Blue

\$n6 color Blue

\$n7 color Blue

\$ns make-lan "\$n0 \$n1 \$n2 \$n3" 100Mb 10ms LL Queue/DropTail Mac/802_3

\$ns make-lan "\$n4 \$n5 \$n6 \$n7" 100Mb 10ms LL Queue/DropTail Mac/802_3

\$ns duplex-link \$n3 \$n4 100Mb 10ms DropTail

\$ns duplex-link-op \$n3 \$n4 color "green"

set err [new ErrorModel]

\$ns lossmodel \$err \$n3 \$n4

\$err set rate_ 0.01

set tcp [new Agent/TCP]

\$tcp set class_ 1

\$ns attach-agent \$n0 \$tcp

set sink [new Agent/TCPSink]

\$ns attach-agent \$n7 \$sink

\$ns connect \$tcp \$sink

set ftp [new Application/FTP]

\$ftp attach-agent \$tcp

```
set udp [new Agent/UDP]
$udp set class_ 2
$ns attach-agent $n0 $udp
```

```
set null [new Agent/Null]
$ns attach-agent $n7 $null
```

```
$ns connect $udp $null
```

```
set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set rate_ 5Mb ;
```

```
proc finish {} {
    global ns nt na
    $ns flush-trace
    close $nt
    close $na
    exec nam 9.nam &
    exit 0
}
```

```
$ns at 0.1 "$ftp start"
$ns at 0.1 "$cbr start"
```

```
$ns at 4.5 "$ftp stop"
$ns at 4.5 "$cbr stop"
```

```
$ns at 5.0 "finish"
```

```
$ns run
```

P9.awk

```
BEGIN{
    pkt=0;
    time=0;
}
{
    if($1=="r" && $4=="7"){
        pkt+=$6;
        time=$2;
    }
}
END{
    printf("Throughput = %fMbps\n",((pkt/time)*(8/1000000)));
}
```

10. Simulate an Ethernet LAN using n nodes and set multiple traffic nodes and plot congestion window for different source/destination

```
set ns [ new Simulator ]
```

```
set tf [open 10.tr w]
```

```
$ns trace-all $tf
```

```
set nf [open 10.nam w]
```

```
$ns namtrace-all $nf
```

```
set n0 [$ns node]
```

```
$n0 color "magenta"
```

```
$n0 label "SRC1"
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
$n2 color "magenta"
```

```
$n2 label "SRC2"
```

```
set n3 [$ns node]
```

```
$n3 color "blue"
```

```
$n3 label "DEST2"
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
$n5 color "blue"
```

```
$n5 label "DEST1"
```

```
$ns make-lan "$n0 $n1 $n2 $n3 $n4 $n5" 100Mb 100ms LL Queue/DropTail Mac/802_3
```

```
$ns duplex-link $n4 $n5 1Mb 1ms DropTail
```

```
set tcp1 [new Agent/TCP]
```

```
$ns attach-agent $n0 $tcp1
```

```
set ftp1 [new Application/FTP]
```

```
$ftp1 attach-agent $tcp1
```

```
$ftp1 set packetSize_ 500
```

```
$ftp1 set interval_ 0.0001
```

```
set sink5 [new Agent/TCPSink]
```

```
$ns attach-agent $n5 $sink5
```

```
$ns connect $tcp1 $sink5
```

```
set tcp2 [new Agent/TCP]
```

```
$ns attach-agent $n2 $tcp2
```

```
set ftp2 [new Application/FTP]
```



```
$ftp2 attach-agent $tcp2
$ftp2 set packetSize_ 600
$ftp2 set interval_ 0.001
```

```
set sink3 [new Agent/TCPSink]
$ns attach-agent $n3 $sink3
$ns connect $tcp2 $sink3
```

```
set file1 [open file1.tr w]
$tcp1 trace cwnd_
$tcp1 attach $file1
```

```
set file2 [open file2.tr w]
$tcp2 trace cwnd_
$tcp2 attach $file2
```

```
proc finish {} {
global ns nf tf
$ns flush-trace
close $tf
close $nf
exec nam 10.nam &
exit 0
}
```

```
$ns at 0.1 "$ftp1 start"
$ns at 5 "$ftp1 stop"
$ns at 7 "$ftp1 start"
$ns at 0.2 "$ftp2 start"
$ns at 8 "$ftp2 stop"
$ns at 14 "$ftp1 stop"
$ns at 10 "$ftp2 start"
```

```
$ns at 15 "$ftp2 stop"
```

```
$ns at 16 "finish"
```

```
$ns run
```

P10.awk

```
BEGIN{
```

```
}
```

```
{
```

```
    if($6=="cwnd_")
```

```
        printf("%f\t%f\n", $1, $7);
```

```
}
```

```
END{
```

```
}
```
